

Two-Output Digital Power Supply

1. Features

- Stable power supply: 12 ± 0.8 V and 5 ± 0.15 V
- Low input voltage required: 3.8 V
- Accurate output voltage
- Wide supply voltage: 2.6 V to 30 V
- Low noise: ripple voltage lower than 140 mVp-p
- Light mass and small volume

2. Applications

- Digital power supply subsystem
- Laboratory testing equipment
- Embedded system
- Robotics power supply
- Electronics design prototyping

3. Description

This board is a digital power supply containing two DC-to-DC converters and current limits which allows users to generate both 12V and 5V outputs with up to 20mA current. It can work solidly in variable conditions without damage, with -2.6V to 6.0V supply and over temperature from 0 to 70 °C.

4. Device information

NAME	BODY SIZE	WEIGHT
5V and 12V DPS	45mm x 45mm	8.00g

5. Simplified Schematics

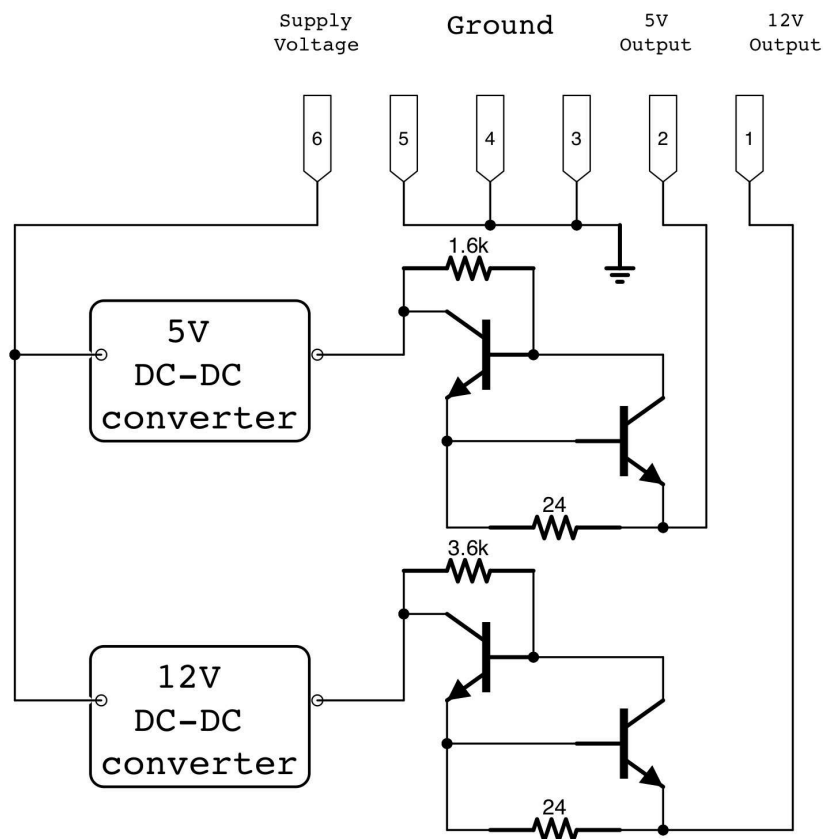


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6. Pin layout

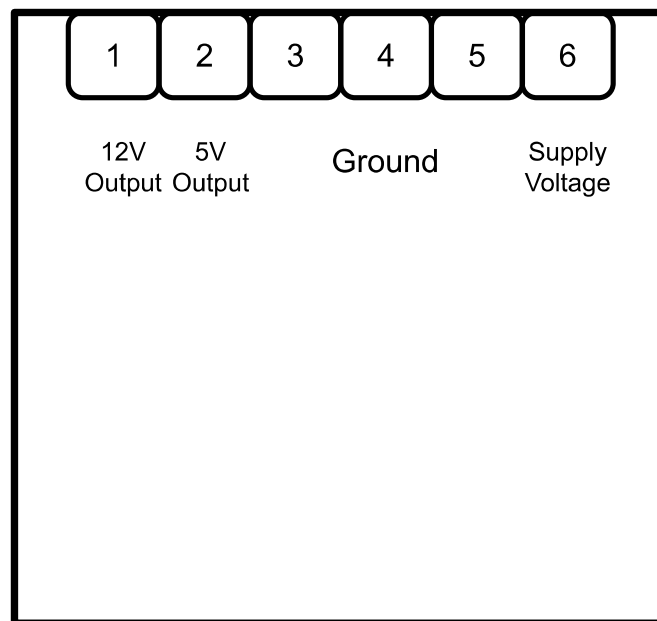


Table 6-1.Pin Functions:xxxNAME

PIN		I/O	DESCRIPTION
NAME	NO.		
IN+	6	I	Positive input
OUT N1	1	O	12V output
OUT N2	2	O	5V output
IN-	5	I	Negative input (ground)
GND N1	3	-	Ground
GND N2	4	-	Ground

7. Specification

7.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)

PARAMETER		Min	Max	Unit
V _{cc}	Supply voltage		30	V
T _{stg}	Storage temperature range	-65	150	°C
I _{cc}	Supply current		1.5	A

- 1) Stresses beyond the listed Absolute Maximum Ratings may cause damage to the device.

7.2 Recommended Operating Conditions

PARAMETER		Min	TYP	Max	Unit
V _{CC}	Supply voltage	2.6	3.8	30	V
T _A	Operating free-air temperature	0	27	70	°C

- 1) Stresses beyond the listed Recommended Operating Conditions may cause inaccurate output.

7.3 Electrical Characteristics- 5V output circuit

PARAMETER	TEST CONDITIONS	T_A				UNIT
			MIN	TYP	MAX	
Input current	Vin = 3.8V, RL1 = 250Ω	27°C	50.000			mA
Output current	Vin = 3.8V, RL1 = 250Ω	27°C	19.954			mA
Output voltage	Vin = 3.8V, RL1 = 250Ω	27°C	4.952	5.008	5.058	V
Efficiency	Vin = 3.8V, RL1 = 250Ω	27°C	52.59%			-
Ripple voltage	Vin = 3.8V, RL1 = 250Ω	27°C	123.1			mVp-p
Ripple voltage without load	Vin = 3.8V	27°C	10.7			mVp-p
Temperature coefficient of output voltage	Vin = 3.8V, RL1 = 250Ω	0°C-70°C	3.9			mV/°C
Short-circuit output current	Vin = 3.8V, RL1 = 250Ω	27°C	28.2			mA
Output voltage variation	Vin = 3.8V, tolerance = 1%	27°C	0.536	7.97		mV
Efficiency variation	Vin = 3.8V, tolerance = 1%	27°C	0.011	0.033		%
Output impedance		27°C	390			kΩ

7.4 Electrical Characteristics– 12V output circuit

PARAMETER	TEST CONDITIONS	T _A				UNIT
			MIN	TYP	MAX	
Input current	V _{in} = 3.8V, RL2 = 600Ω	27°C	101.500			mA
Output current	V _{in} = 3.8V, RL2 = 600Ω	27°C	19.851			mA

Output voltage	RL2 = 600Ω	27°C	11.956	12.088	12.211	V
Efficiency	Vin = 3.8V, RL2 = 600Ω	27°C	62.21%			-
Ripple voltage	Vin = 3.8V, RL2 = 600Ω	27°C	138.9			mVp-p
Ripple voltage without load	Vin = 3.8V	27°C	52.6			mVp-p
Temperature coefficient of output voltage	Vin = 3.8V, RL2 = 600Ω	0°C-70°C	5.3			mV/°C
Short-circuit output current	Vin = 3.8V, RL2 = 600Ω	27°C	27.95			mA
Output voltage variation	Vin = 3.8V, tolerance = 1%	27°C	1.3		25.01	mV
Efficiency variation	Vin = 3.8V, tolerance = 1%	27°C	0		0.067	%
Output impedance		27°C	400			kΩ

7.5 Electrical Characteristics–Total device

PARAMETER	TEST CONDITIONS	T_A				UNIT
			MIN	TYP	MAX	
Efficiency	Vin = 3.8V, RL1 = 250Ω, RL2 = 600Ω	27°C	62.11%			-
Reference voltage, Vref		27°C	0			V
5V output voltage	Vin = 3.8V, I _{load} ≤ 20mA	27°C	5.0	5.009	5.135	V
12V output voltage	Vin = 3.8V, I _{load} ≤ 20mA	27°C	12.01	12.092	12.58	V
Quiescent Current	Vin = 3.8V	27°C	10			mA
Output noise voltage	Vin = 3.8V, RL1 = 250Ω, RL2 = 600Ω	27°C	10.6			μV
Supply current, I _{cc}	Vin = 3.8V, RL1 = 250Ω, RL2 = 600Ω	27°C	144.5			mA
Input impedance		27°C	958			Ω

7.6 Typical Characteristics (plots)

At $T = 27^{\circ}\text{C}$

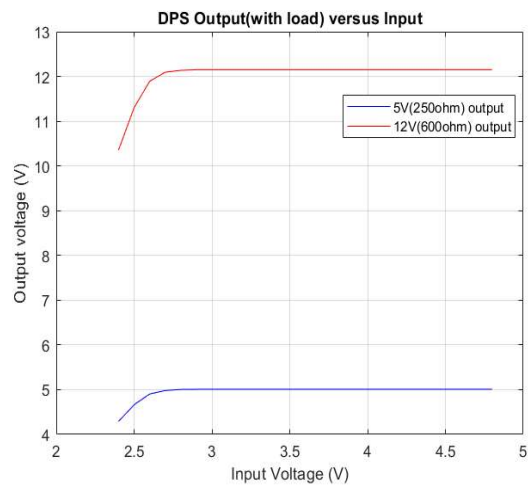


Figure 1: DPS Output voltage versus input with load

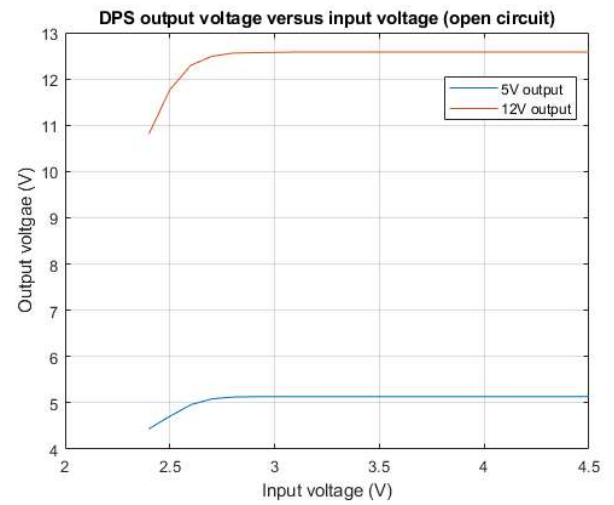


Figure 2: DPS Output voltage versus input without load

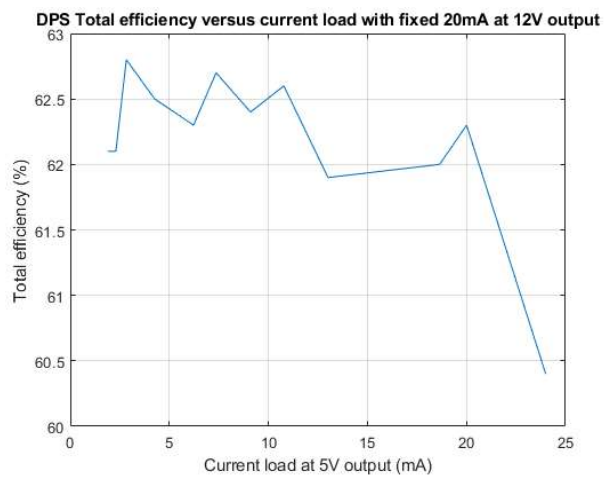


Figure 3: total efficiency versus 5V current load

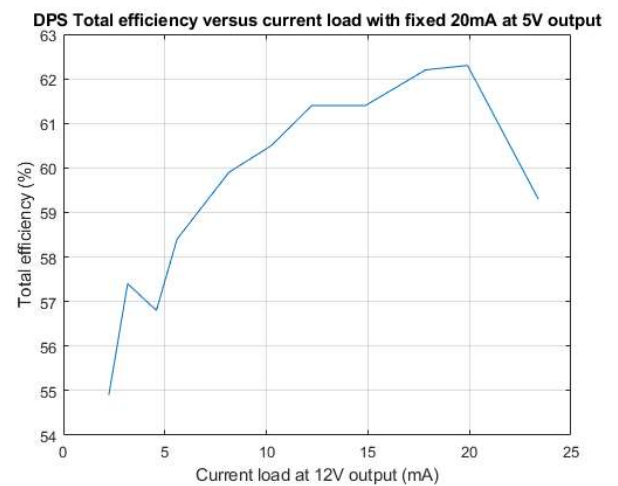


Figure 4: total efficiency versus 12V current load

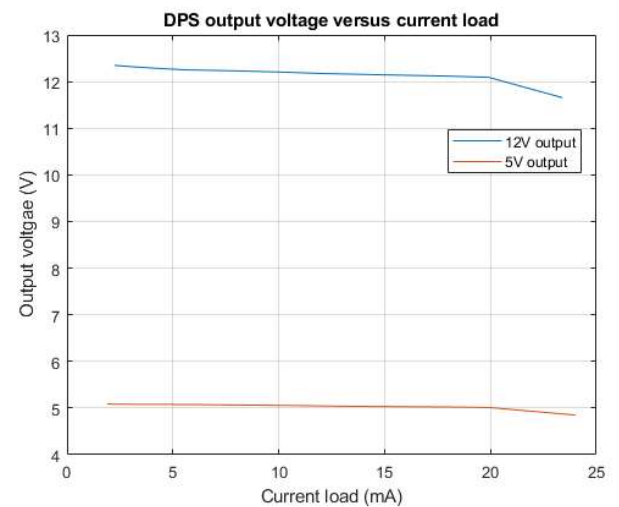
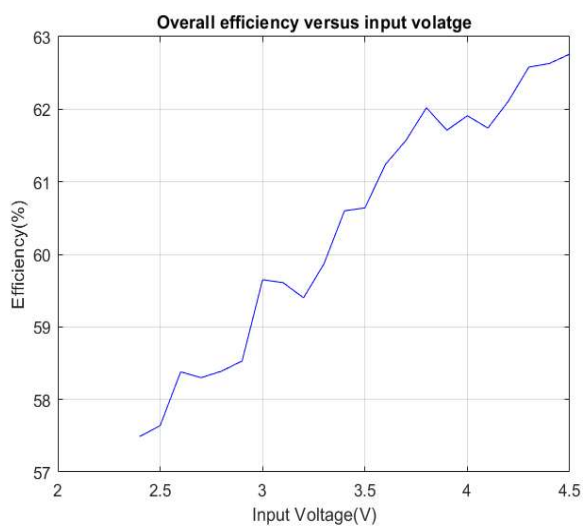


Figure 5: Total efficiency versus input voltage

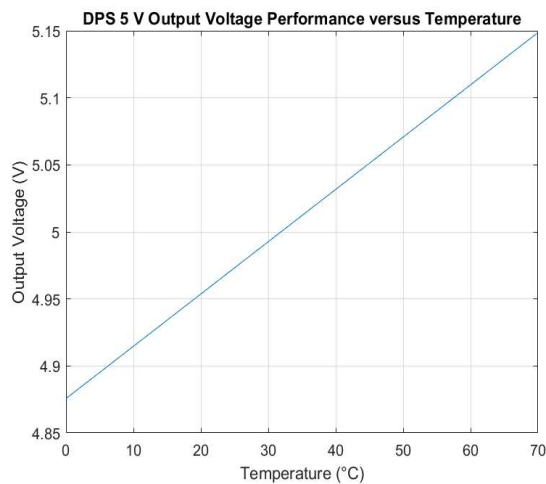


Figure 7: 5 V Output voltage versus Temperature
(Results based on Multisim Simulation)

Figure 6: Output voltage versus current load

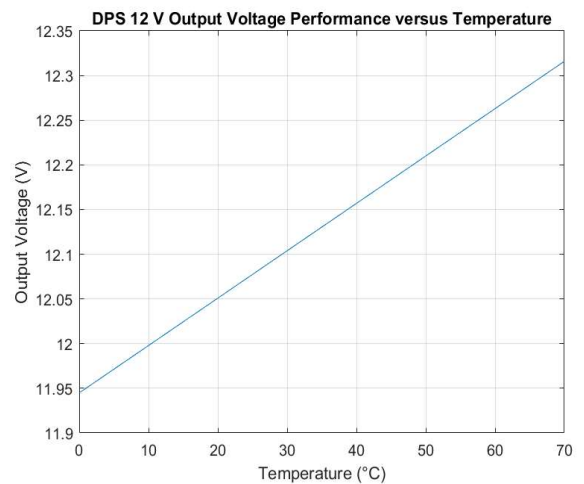


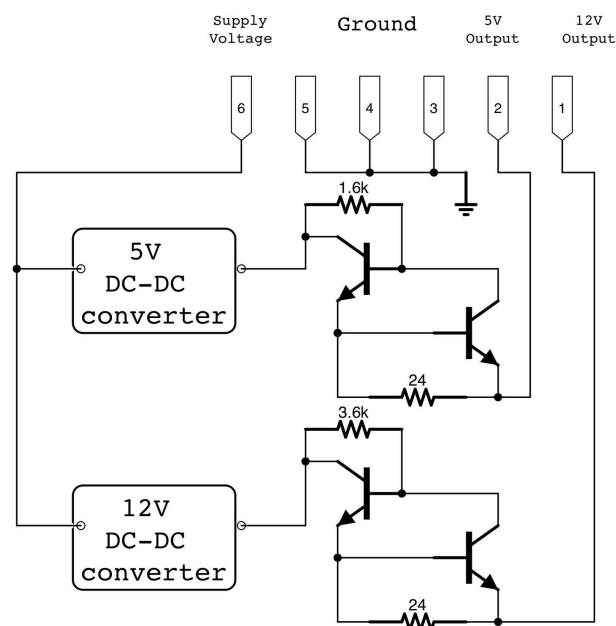
Figure 8: 12 V Output voltage versus Temperature
(Results based on Multisim Simulation)

8. Detailed Description

8.1 Overview

This board is a digital power supply containing two DC-to-DC converters and current limits which allows users to generate both 12V and 5V outputs with up to 20mA current. The converters mainly use MC33063AD chips and the current limits are made up of BC817-40W NPN BJTs and 0805 resistors. The total efficiency of it is over 60% with two outputs. The board can work solidly in variable conditions without damage, with -2.6V to 6.0V supply and over temperature from 0 to 70 °C. It cannot be destroyed at -40°C and available to be used or stored in many experimental situations.

8.2 Functional Block Diagram



8.3 Feature Description

- Stable power supply: 12 ± 0.8 V and 5 ± 0.15 V

- Low input voltage required: 3.8 V
- Accurate output voltage
- Wide supply voltage: 2.6 V to 30 V
- Low noise: ripple voltage lower than 140 mVp-p
- Light mass and small volume

9. Application and implementation

A typical application for the board is a boost converter. This converter can step up input voltage to higher specific values with two separate output pins. At the same time, it limits the current to a given value.

10. Power Supply Recommendation

CAUTION

Supply voltages larger than 30 V can permanently damage the device.

11. Schematics and PCB Layout

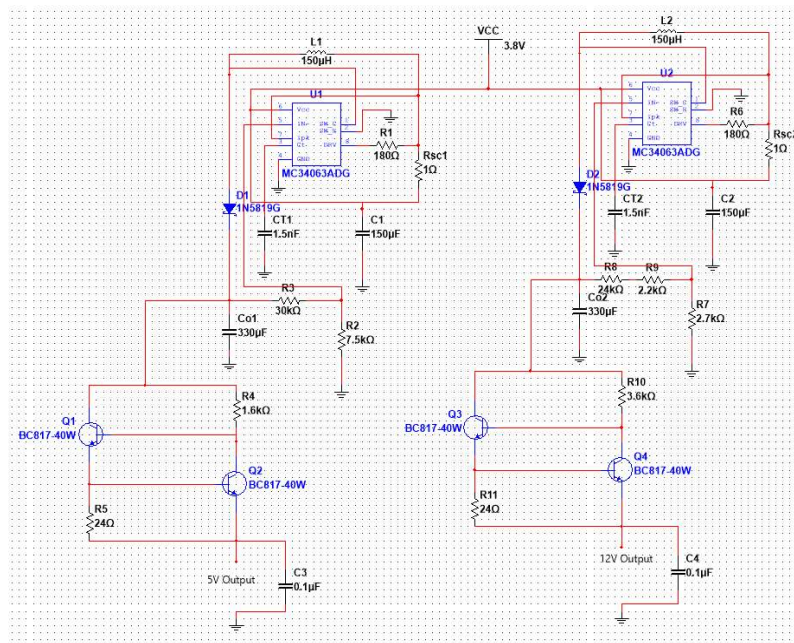


Figure 9: Schematics layout

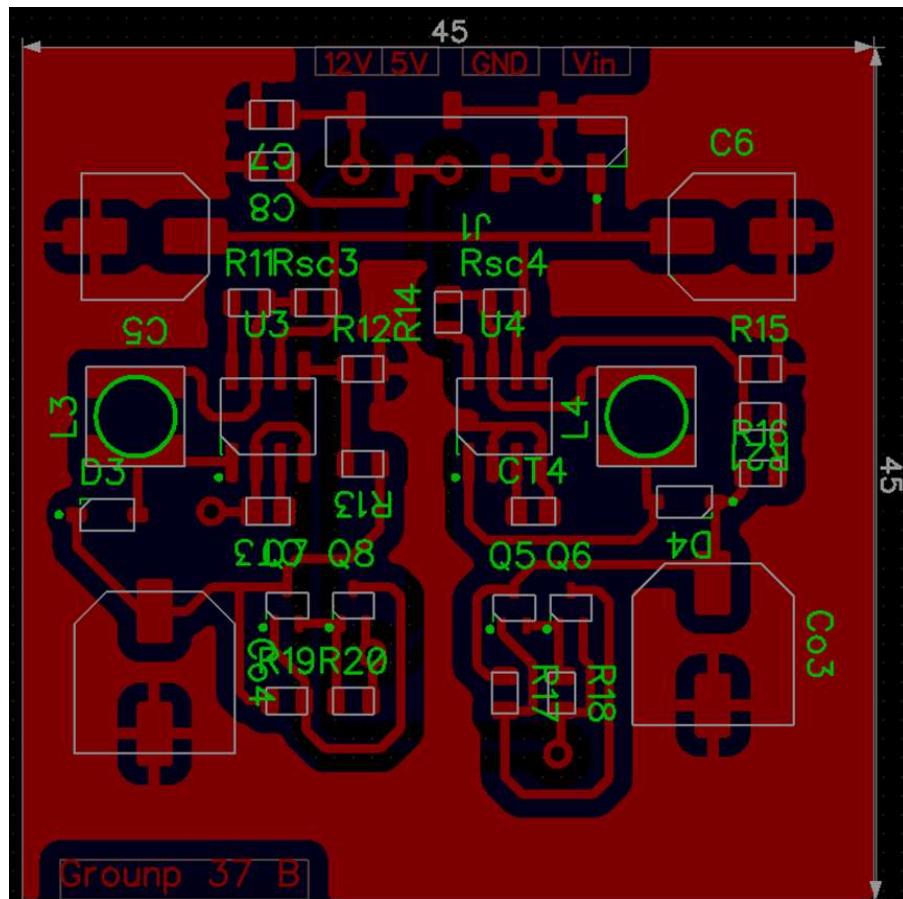


Figure 10: PCB front layout

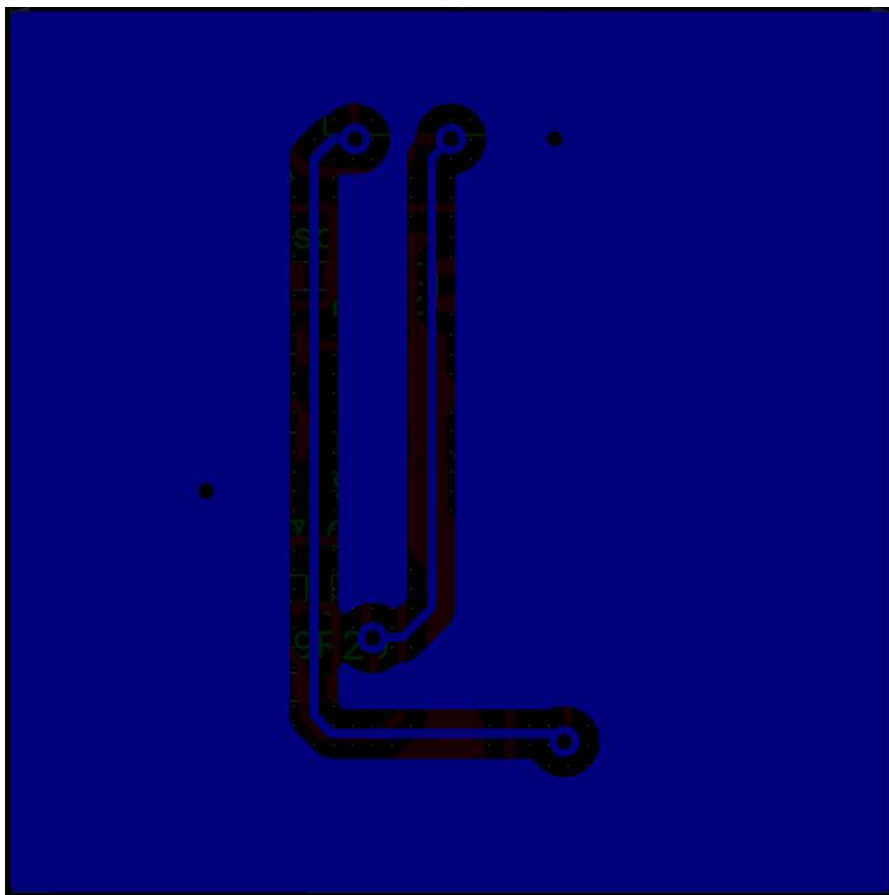


Figure 11: PCB back layout

12. Component List

Table 12-1. Component List

Component	Value / model	Number
0805Resistor	1	2
0805Resistor	24	2
0805Resistor	180	2
0805Resistor	1.6k	1
0805Resistor	2.2k	1
0805Resistor	2.7k	1
0805Resistor	3.6k	1
0805Resistor	7.5k	1
0805Resistor	24k	1
0805Resistor	30k	1
0805Capacitor	1.5n	2
0805Capacitor	100n	2
Electrolytic Capacitors	MCVVT016M151EA6L 150u	2
Electrolytic Capacitors	MCVVT016M331FB3L 330u	2
Inductor	SRR4528A 150u	2
Rectifier	1N5819HW	2
NPN BJT	BC817-40W	4
ICs	MC33063AD	2

Connector	M20-8770646	1
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13. Related links

This table shows the components used and necessary technical resources of them.

Table 13-1. Related Links

PART NUMBER	TYPE	TECHNICAL DOCUMENTS
BC817-40W	NPN BJT	45 V, 500 mA NPN general-purpose transistors (mouser.co.uk)
MC33063AD	IC	MC3x063A 1.5-A Peak Boost/Buck/Inverting Switching Regulators datasheet (Rev. N)
1N5819HW	Rectifier	1N5819HW (mouser.co.uk)
SRR4528A	Inductor	srr4528a_after.pdf (bourns.com)
MCVVT016M151EA6L 150u	Electrolytic capacitors	2151198.pdf (farnell.com)
MCVVT016M151EA6L 330u	Electrolytic capacitors	3486024.pdf (farnell.com)
HARWIN M20-8770646	Header	Component Specification C003 - 2.54mm pitch Jumper Sockets (farnell.com)